

ECO5050 AI Applications and Techniques in Economics

Lecture 1: Loss functions and forecast optimality

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Fall 2025

Outline

- 1 Loss Functions
- 2 Optimal Point Forecasts
- 3 Classical Approach

The Forecasting Problem

- Forecasting can be viewed as the process involved in providing information on future values of one or more variables of interest.
- Important elements of deriving the forecast
 - ① Loss function
 - ② Family of forecasting models: parametric, semiparametric, nonparametric
 - ③ Type of information to report for the outcome: point, interval, density
- Notation
 - ▶ Y is the random variable describing the outcome of interest
 - ▶ Z is the random variable with an outcome that is the collection of all relevant information at the time of the forecast
 - ▶ We focus on point forecasts $f(z, \theta)$ which is a function of the available data z and a set of unknown parameters θ

Loss Functions

- No forecast is going to be always correct, so we need a specification of how costly different mistakes are to guide the procedure.
- The loss function $L(f, y, z)$ describes in relative terms how costly it is to use an imperfect forecast f , given the outcome y , and other observed data z .

$$L : \mathcal{F} \times \mathcal{Y} \times \mathcal{Z} \mapsto \mathcal{L} \subseteq \mathbb{R}$$

- ▶ Usually $\mathcal{L} \subseteq \mathbb{R}_+$, the set of nonnegative real numbers
- ▶ e.g. A policy maker's inflation forecast loss function may depend on the unemployment rate
- A common assumption for loss functions is that loss is minimized when the forecast is equal to the outcome:

$$\min_f L(f, y, z) = L(y, y, z)$$

Common Properties of Loss Functions

- For simplicity we will drop the explicit dependence on z and focus on simpler loss functions $L(f, y)$.
- We often impose that the loss function is nondecreasing as f moves further away from y ; bigger forecast errors imply bigger losses.
- For loss functions that only depend on the forecast error $e = y - f$, Granger (1999) summarized these requirements:
 - 1 $L(0) = 0$ (minimal loss of 0)
 - 2 $L(e) \geq 0$ for all e
 - 3 $L(e)$ is nonincreasing in e for $e < 0$ and nondecreasing in e for $e > 0$

$$L(e_1) \leq L(e_2) \text{ if } e_2 < e_1 < 0, \quad L(e_1) \leq L(e_2) \text{ if } 0 < e_1 < e_2$$

Common Properties of Loss Functions

- Other properties of loss functions: homogeneity, symmetry, differentiability, boundness
 - ▶ Homogeneity: $L(af, ay) = h(a)L(f, y)$ for some positive function $h(a)$
 - ▶ Symmetry: for all a , $L(y - a, y) = L(y + a, y)$
 - ▶ Differentiability of $L(f, y)$ with respect to f is a regularity condition that helps simplify the search for optimal forecasts.
 - ▶ The loss function L may be bounded or unbounded, but unbounded L can create technical problems as the expected loss may not exist.
- The most commonly used loss function is the quadratic (squared error) loss:

$$L(e) = ae^2 = a(y - f)^2, \quad a > 0$$

Specific Loss Functions

Loss that depends only on forecast errors: symmetric loss

- **Squared error loss:**

$$L(e) = ae^2, \quad a > 0$$

satisfies the three Granger properties + symmetric, homogeneous, differentiable everywhere, unbounded from above

- **Absolute error loss**

$$L(e) = a|e|, \quad a > 0$$

satisfies the three Granger properties + symmetric, homogeneous, differentiable everywhere except at 0, unbounded from above

Specific Loss Functions

Loss that depends only on forecast errors: asymmetric loss

- Piecewise linear/Lin-lin loss: for $a > 0$, $0 < \alpha < 1$

$$L(e) = \begin{cases} -a(1 - \alpha)e & \text{if } e \leq 0 \\ a\alpha e & \text{if } e > 0 \end{cases}$$

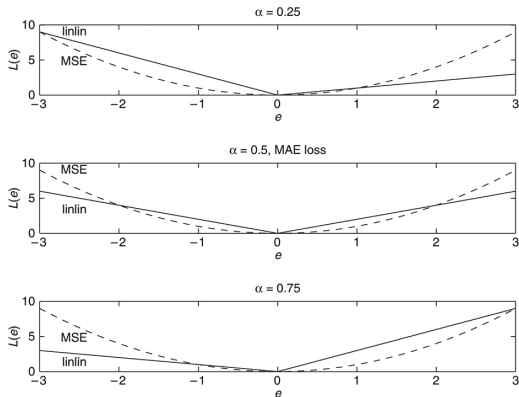
satisfies the three Granger properties + homogeneous for a positive constant a , differentiable everywhere except at 0, unbounded from above.

- Positive forecast errors are assigned a relative weight of α , negative errors get a weight of $1 - \alpha$.
- Corresponds to absolute error loss when $\alpha = 0.5$

Specific Loss Functions

Loss that depends only on forecast errors: asymmetric loss

- Squared error (MSE) loss versus lin-lin loss for different values of the lin-lin asymmetry parameter α



Specific Loss Functions

Loss that depends only on forecast errors: asymmetric loss

- Linear-exponential/Linex loss (Varian, 1975; Zellner, 1986)

$$L(e) = a_1 [\exp(a_2 e) - a_2 e - 1], \quad a_2 \neq 0, a_1 > 0$$

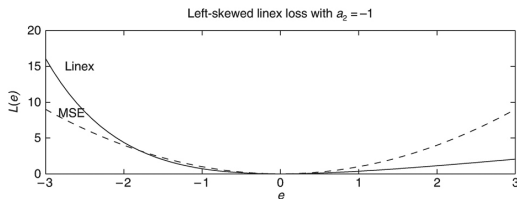
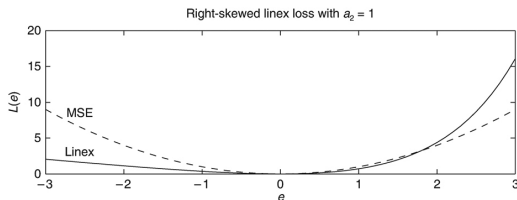
satisfies the three Granger properties + differentiable everywhere, unbounded from above.

- The parameter a_2 controls both the degree and direction of asymmetry.
 - ▶ If $a_2 > 0$, linex loss is approximately linear for $e < 0$ and approximately exponential for $e > 0$.
 - ▶ If $a_2 < 0$, linex loss is approximately exponential for $e < 0$ and approximately linear for $e > 0$.
- Example: Budget forecasts (Artis and Marcellino, 2001), survey forecasts of inflation (Capistrán and Timmermann, 2009)

Specific Loss Functions

Loss that depends only on forecast errors: asymmetric loss

- MSE loss versus Linex loss for different values of the parameter a_2



Optimal Point Forecasts

- The forecaster's objective is to use data (outcomes of the random variable) Z to predict the value of the random variable Y
 $\Rightarrow$ Find the decision rule, the forecast $f(Z)$, that is close to the outcome Y on average.
 - ▶ Assume that a joint distribution for $\{Y, Z\}$ exists, $p_Y(y|z, \theta)p_Z(z|\theta)$
 - ▶ The forecast $f(Z, \theta)$ depends on a set of parameters θ and data Z
- Classical forecasting methods for choosing $f(Z, \theta)$ attempt to control and minimize the risk

$$\begin{aligned} R(\theta, f) &= E_{Y,Z} [L(f(Z, \theta), Y)] \\ &= \int_z \int_y L(f(z), y) p_Y(y|z, \theta) p_Z(z|\theta) dy dz \end{aligned}$$

Optimal Point Forecasts

- The classical approach focuses on evaluating the inner integral, the expected loss conditional on Z

$$E_Y [L(f(Z, \theta), Y)|Z] = \int_Y L(f(z, \theta), y) p_Y(y|z, \theta) dy$$

- When the expected loss exists, the optimal point forecast $f^* \in \mathcal{F}$ satisfies

$$f^*(z) = \arg \min_{f \in \mathcal{F}} \int_Y L(f(z, \theta), y) p_Y(y|z, \theta) dy \quad (1)$$

- Closed-form solutions to (1) can be established for many commonly used loss functions.

Optimal Point Forecasts

Squared forecast error loss

- Suppose we want to obtain $f^*(z_t, \theta)$, the one-step-ahead forecast of y_{t+1} under the MSE loss

$$E[(Y_{t+1} - f(Z_t, \theta))^2 | Z_t, \theta] = \int_y (y_{t+1} - f(z_t, \theta))^2 p_Y(y_{t+1} | z_t, \theta) dy_{t+1}$$

- ▶ The random variable Z_t is defined on the information set $\Omega_t = (Y_t, X_t, Y_{t-1}, X_{t-1} \dots)$
- The first-order condition:

$$\begin{aligned} \frac{d}{df(z_t)} \int_y (y_{t+1} - f(z_t))^2 p_Y(y_{t+1} | z_t) dy_{t+1} \\ = \int_y \frac{\partial}{\partial f(z_t)} (y_{t+1} - f(z_t))^2 p_Y(y_{t+1} | z_t) dy_{t+1} \\ = -2 \int_y (y_{t+1} - f(z_t)) p_Y(y_{t+1} | z_t) dy_{t+1} = 0 \end{aligned}$$

Optimal Point Forecasts

Squared forecast error loss

- The first-order condition:

$$f^*(z_t, \theta) = \int_y y_{t+1} p_Y(y_{t+1}|z_t) dy_{t+1} = E[Y_{t+1}|Z_t = z_t, \theta]$$

- If we wish to minimize the expected square error loss given information at time t , the optimal forecast is the expected value of y_{t+1} given variables we know at time t .
- Note that setting the forecast $f(z)$ to $E[Y_{t+1}|Z_t = z_t, \theta]$ does not guarantee that the outcome y_{t+1} will be close to the forecast, since there is no guarantee that a particular outcome is close to its conditional mean. However, on average y_{t+1} will be close to the conditional mean.

Optimal Point Forecasts

Lin-lin loss

- The lin-lin loss

$$L(e) = \alpha e \mathbf{1}(e > 0) - (1 - \alpha)e \mathbf{1}(e \leq 0), \quad 0 < \alpha < 1$$

$$\mathbf{1}(e > 0) = \begin{cases} 1 & \text{if } e > 0 \\ 0 & \text{otherwise} \end{cases}$$

- The expected loss

$$\begin{aligned} \int_y L(f(z), y) p_Y(y|z, \theta) dy &= \alpha \int_{f(z)}^{\infty} (y - f) p_Y(y|z, \theta) dy \\ &\quad - (1 - \alpha) \int_{-\infty}^{f(z)} (y - f) p_Y(y|z, \theta) dy \end{aligned}$$

Optimal Point Forecasts

Lin-lin loss

- Differentiate the expected loss with respect to $f(z)$ yields the FOC for an optimal forecast

$$\begin{aligned}\alpha (1 - P_{Y|Z}(f^*(z))) + (1 - \alpha)P_{Y|Z}(f^*(z)) &= 0 \\ P_{Y|Z}(f^*(z)) &= \alpha\end{aligned}$$

where $P_{Y|Z}(x) = \int_{-\infty}^x p_Y(y|z, \theta)dy$

- The optimal forecast is $f^*(z) = P_{Y|Z}^{-1}(\alpha)$, which is the quantile function for Y given Z
 - ▶ As α decreases toward 0, the optimal forecast moves further to the left of the tail of the outcome distribution because negative forecast errors (overpredictions) are penalized more heavily than positive errors.

Optimal Point Forecasts

Linex loss

- The Linex loss function is

$$L(f(z), y) = \exp(a(y - f(z))) - a(y - f(z)) - 1$$

and so

$$\int_y L(f(z), y) p_Y(y|z, \theta) dy = \exp(-af(z)) M_{Y|z}(a) - a(E[Y|z] - f(z)) - 1$$

where $M_{Y|z}(t) = E[\exp(Yt)|z]$ is the moment generating function for Y conditional on Z , which we assume exists and $a \neq 0$ is a parameter

- The FOC gives the closed-form solution for the optimal forecast

$$\begin{aligned} -a \exp(-af^*(z)) M_{Y|z}(a) + a &= 0 \\ f^*(z) &= a^{-1} \ln M_{Y|z}(a) \end{aligned}$$

Interpretation of Forecast Optimality

- The optimal forecast minimizes the expected loss, hence it aims to work well on average rather than for a particular value (single draw) of the outcome Y .
- The optimal forecast depends on the conditioning variables used to produce the forecast, Z . Conditioning on different predictor variables generally leads to different optimal forecasts given those predictor variables.
- The optimization problem is conducted with respect to a class of functions $\mathcal{F}$, which is often too wide to suggest the form of a reasonable forecast. We might limit the set of models to specific parametric models rather than extending it to a wide class of functions.
 - ▶ Example: MSE loss and linear projections

Interpretation of Forecast Optimality

Example: MSE loss and linear projections

- Under MSE loss the optimal forecast takes the form $f^*(z) = E[Y|z]$.
- Computing the conditional mean requires knowledge about the true relationship between Y and Z .
- Suppose we restrict the class of functions to be a linear function of Z
- The **linear projection** of Y on Z is the forecast that produces the smallest MSE among the class of linear forecast models:

$$\hat{E}[Y|Z] \equiv Z'\beta^*$$

where β^* satisfies the orthogonality condition

$$E[Z(Y - Z'\beta^*)] = \mathbf{0}$$

Interpretation of Forecast Optimality

Example: MSE loss and linear projections

- Linear projections are a subset of all possible models, so for $f(z) = z'\beta^*$,

$$MSE(f(z)) \geq MSE(E[Y|z]).$$

In general linear projections are suboptimal in population.

- The linear projection $f(z) = \hat{E}[Y|z]$ is the optimal *linear* forecast. Within the class of linear models, the linear projection is the optimal forecast.

Classical Approach

- In practice the parameters θ of the forecast model $f(z, \theta)$ are unknown.
- The two-step approach under MSE loss
 - ① Specify the forecast model $f(z, \theta)$ that is equal to the conditional mean of Y given $Z = z$
 - ★ If $f(z, \theta)$ is equal to $E[Y|z]$, the model is correctly specified. If not, $f(z, \theta)$ is an approximation of $E[Y|z]$.
 - ② Plug-in an estimate $\hat{\theta}$ to generate a forecast $f(z, \hat{\theta})$
- Parameter estimation induces additional randomness and increases the risk; the estimated forecast model $f(z, \hat{\theta})$ will never obtain the optimality property of $f^*(z, \theta)$

Classical approach

Example: MSE loss given the observed history of the outcome

- Consider forecasting Y_{T+1} under MSE loss given $Z_T = \{Y_t\}_{t=1}^T$, where $Y_t \stackrel{i.i.d.}{\sim} N(\mu, \sigma^2)$.
- The optimal forecast is $f^*(Z_T) = E[Y_{T+1}|Z_T] = \mu$, which produces a risk $R(\theta, f^*) = E[(Y_{T+1} - \mu)^2] = \sigma^2$.
- When μ is unknown, we can plug-in the estimator $\hat{\mu} = \bar{Y}_T \sim N(\mu, T^{-1}\sigma^2)$

$$\begin{aligned} R(\theta, \hat{\mu}) &= E[(Y_{T+1} - \bar{Y}_T)^2] \\ &= E[(Y_{T+1} - \mu)^2] + E[(\bar{Y}_T - \mu)^2] \\ &= \sigma^2(1 + T^{-1}) \end{aligned}$$

- This is strictly greater than the risk of the optimal forecast, σ^2

Classical approach

Example: MSE loss for linear forecasting model

- Note that the conditional expected loss $E[L(f(Z_T), Y_{T+1})|Z_T]$ is a function of Z_T ; whereas the risk is $E[L(f(Z_T), Y_{T+1})]$, the expected loss over Y_{T+1} and Z_T .
- Consider a linear forecasting model $y_{t+1} = \beta'x_t + \varepsilon_{t+1}$ where $\varepsilon_{t+1} \stackrel{i.i.d.}{\sim} (0, \sigma^2)$ is independent of $Z_T = (x'_0, \dots, x'_T)'$.
- Forecast is $f(Z_T) = \hat{\beta}'_T x_T$ where

$$\hat{\beta}_T = \left(T^{-1} \sum_{t=0}^{T-1} x_t x'_t \right)^{-1} \left(T^{-1} \sum_{t=0}^{T-1} x_t y_{t+1} \right)$$

Classical approach

Example: MSE loss for linear forecasting model

- The conditional expected loss is

$$\begin{aligned} E \left[(y_{T+1} - \hat{\beta}'_T x_T)^2 \middle| Z_T \right] &= E \left[\varepsilon_{T+1}^2 + (\hat{\beta}_T - \beta)' x_T x_T' (\hat{\beta}_T - \beta) \middle| Z_T \right] \\ &= \sigma^2 \left(1 + T^{-1} x_T' \left(T^{-1} \sum_{t=0}^{T-1} x_t x_t' \right)^{-1} x_T \right) \end{aligned}$$

which depends on all values of x up to time T .

- The unconditional expected loss (risk) is obtained by integrating over Z_T

$$\begin{aligned} R(\theta, \hat{\beta}'_T x_T) &= E \left[\sigma^2 \left(1 + T^{-1} x_T' \left(T^{-1} \sum_{t=0}^{T-1} x_t x_t' \right)^{-1} x_T \right) \right] \\ &\approx \sigma^2 (1 + T^{-1} K) \end{aligned}$$

where $K = \dim(x_t)$.

Classical approach

- Risk functions that assume a correct specification $f(z, \theta)$ and a known value of θ can be considered infeasible lower bounds on risk.
- When considering optimality of a forecast, parameter estimation error should be taken into account.
- A direct consequence of considering the estimation error is that even with a correct model specification there is no single optimal forecast procedure that minimizes risk for all possible values of θ .
- A forecast method belongs to a family of optimal procedures if there exists some region of the parameter space, θ , for which no other forecast method has lower risk.

Classical approach

MSE loss for shrinkage estimator

- Consider the previous example of forecasting Y_{T+1} given $Z_T = \{Y_t\}_{t=1}^T$, where $Y_t \stackrel{i.i.d.}{\sim} N(\mu, \sigma^2)$.
- Suppose we use the shrinkage estimator as the forecast, $f(Z_T) = \gamma \bar{Y}_T$, where $0 < \gamma < 1$

$$\begin{aligned} R(\theta, \gamma \bar{Y}_T) &= E[(Y_{T+1} - \gamma \bar{Y}_T)^2] \\ &= E[(Y_{T+1} - \mu)^2] + \gamma^2 E[(\bar{Y}_T - \mu)^2] + (1 - \gamma)^2 \mu^2 \\ &= \sigma^2(1 + \gamma^2 T^{-1}) + (1 - \gamma)^2 \mu^2 \end{aligned}$$

- For μ in the range $\pm \sqrt{T^{-1}\sigma^2(1 + \gamma)/(1 - \gamma)}$ the forecast $\gamma \bar{Y}_T$ has lower risk than the sample mean $\bar{Y}_T$.